Ryerson University

Department of Electrical, Computer, & Biomedical Engineering
Faculty of Engineering & Architectural Science

Course Title	**ELE734**
Course Name	**Low Power Digital Circuit**
Semester/Year	**F2025**

Instructor	**Dr Andy Ge**

Lab Number	**Lab 4**
Lab Name	**CMOS Inverters**

Submission Date	**2025-11-27**
Due Date	**2025-12-4**

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*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab reports your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a “0” on the work, an “F” in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: http://www.ryerson.ca/senate/current/pol60.pdf

Introduction

In this lab, a 1-bit full adder with inputs A, B, and Cin and outputs S and Cout is implemented. The design is sized using the logical effort method to drive a load on the sum output. The adder's performance is then checked using by holding B and Cin at 0V and 1.8V, and the worst-case delay is verified through simulation.

Results and Observations

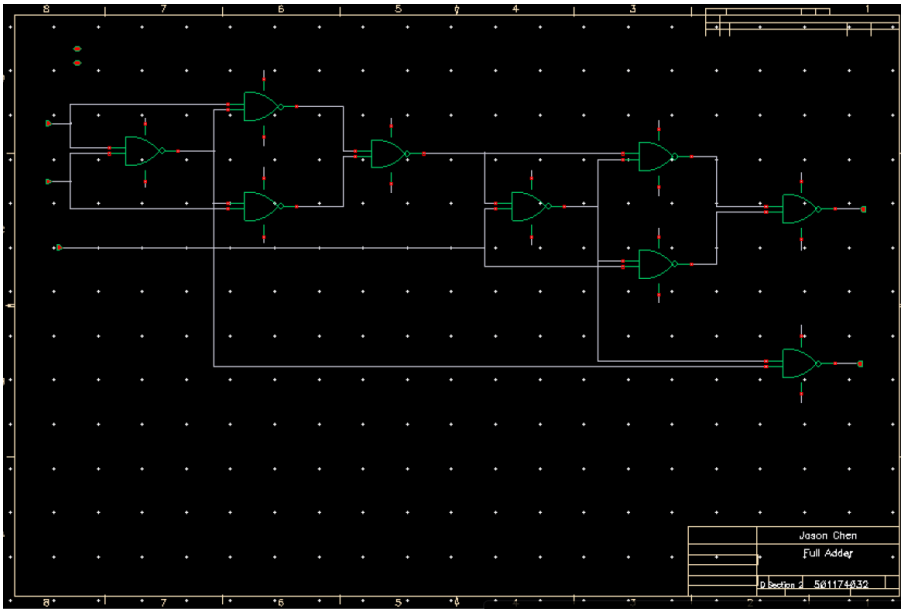


Figure 1.0: Full Adder 2 Input NAND Gate Implementation Schematic

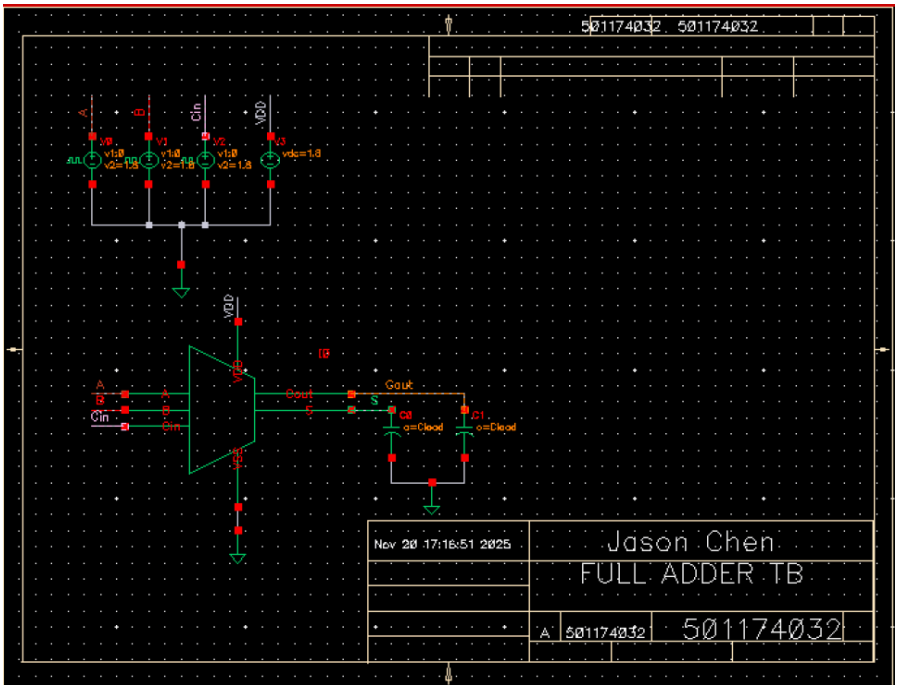


Figure 1.1: Full Adder Testbench

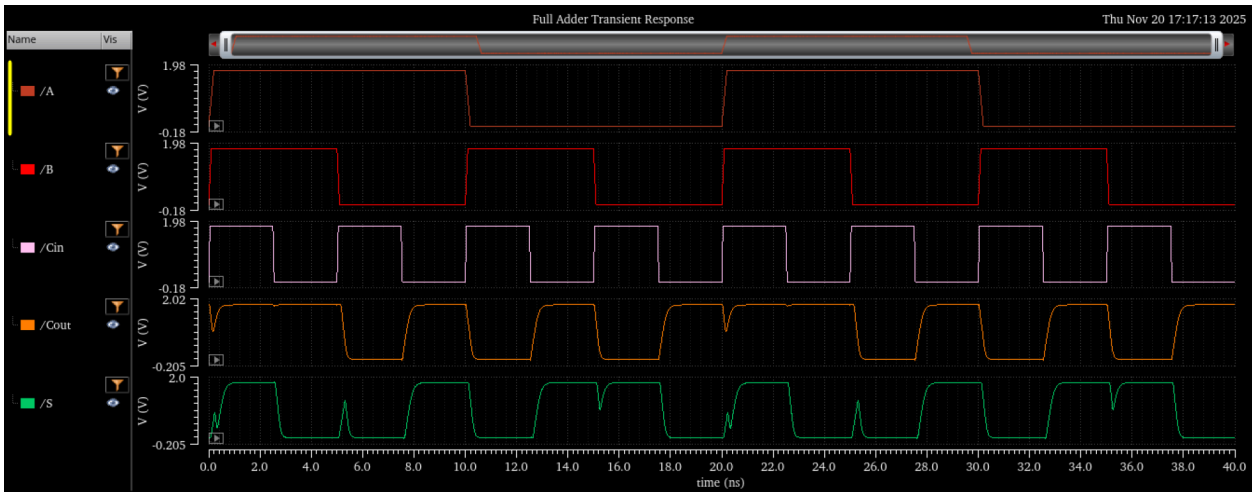


Figure 1.2: Transient Response

6	Cout Rising Delay	337.116p	☑	☐	
7	Cout Falling Delay	312.645p	☑	☐	
8	S Falling Delay	333.895p	☑	☐	
9	S Rising Delay	400.186p	☑	☐	

Figure 1.3: Rise Fall Delay

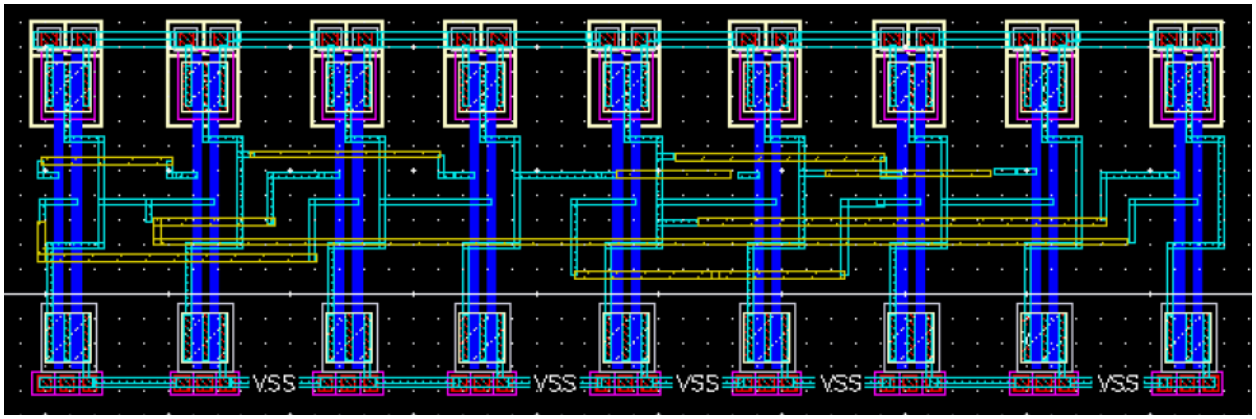


Figure 2.0: Full Adder NAND2 Implementation Layout

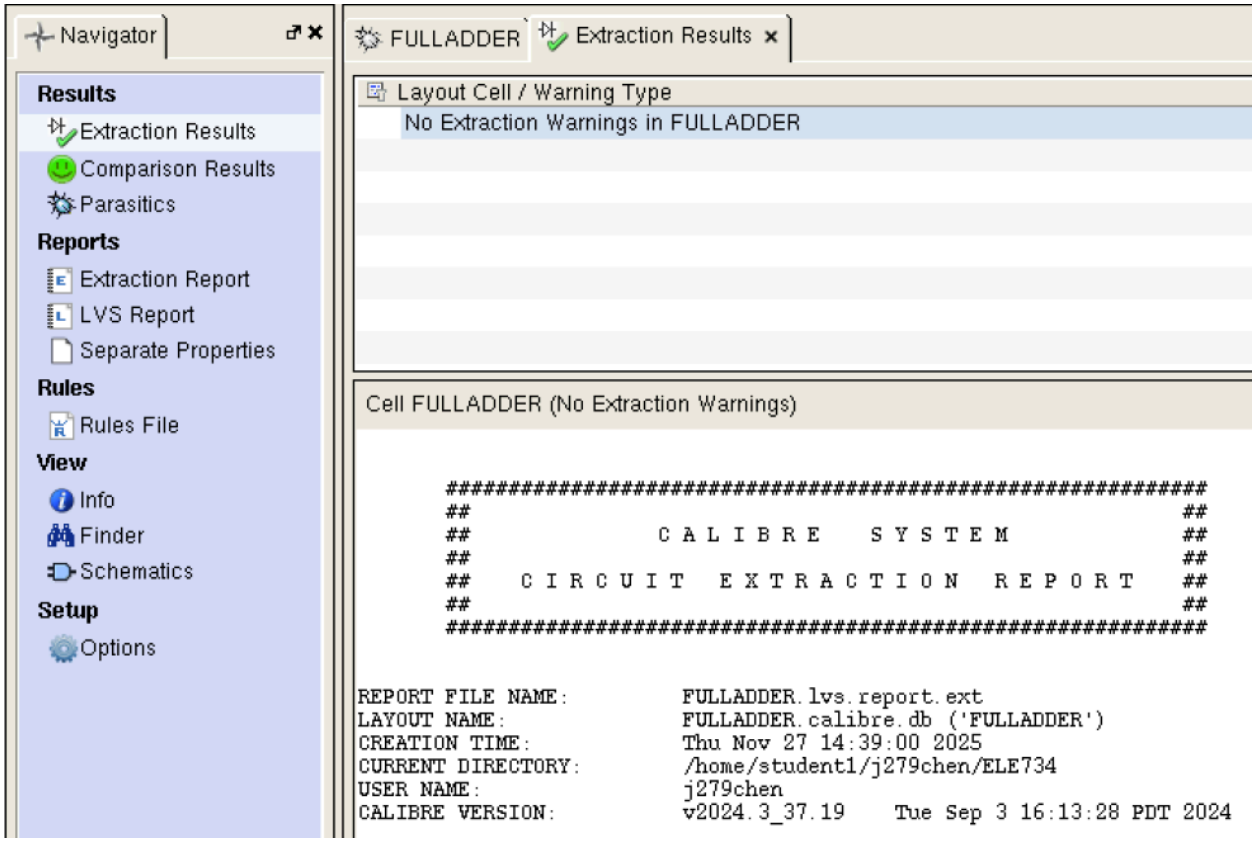


Figure 2.1: DRC Check

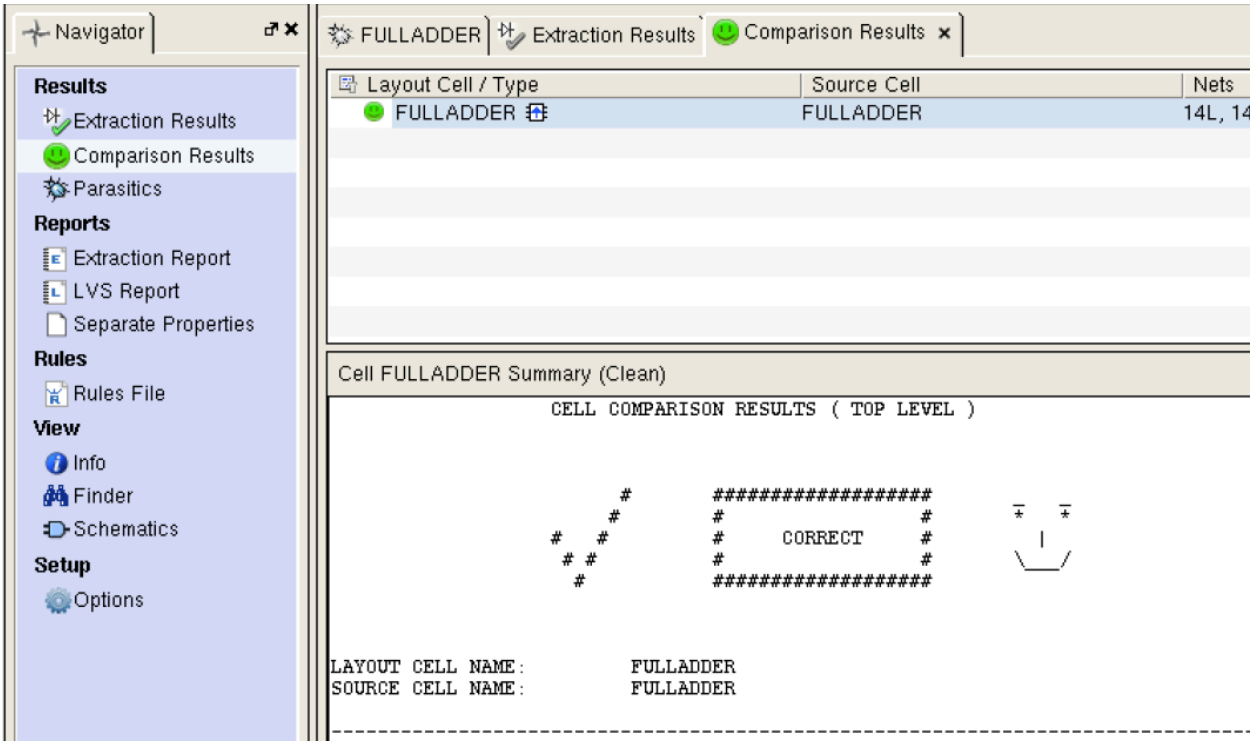


Figure 2.2: LVS Check

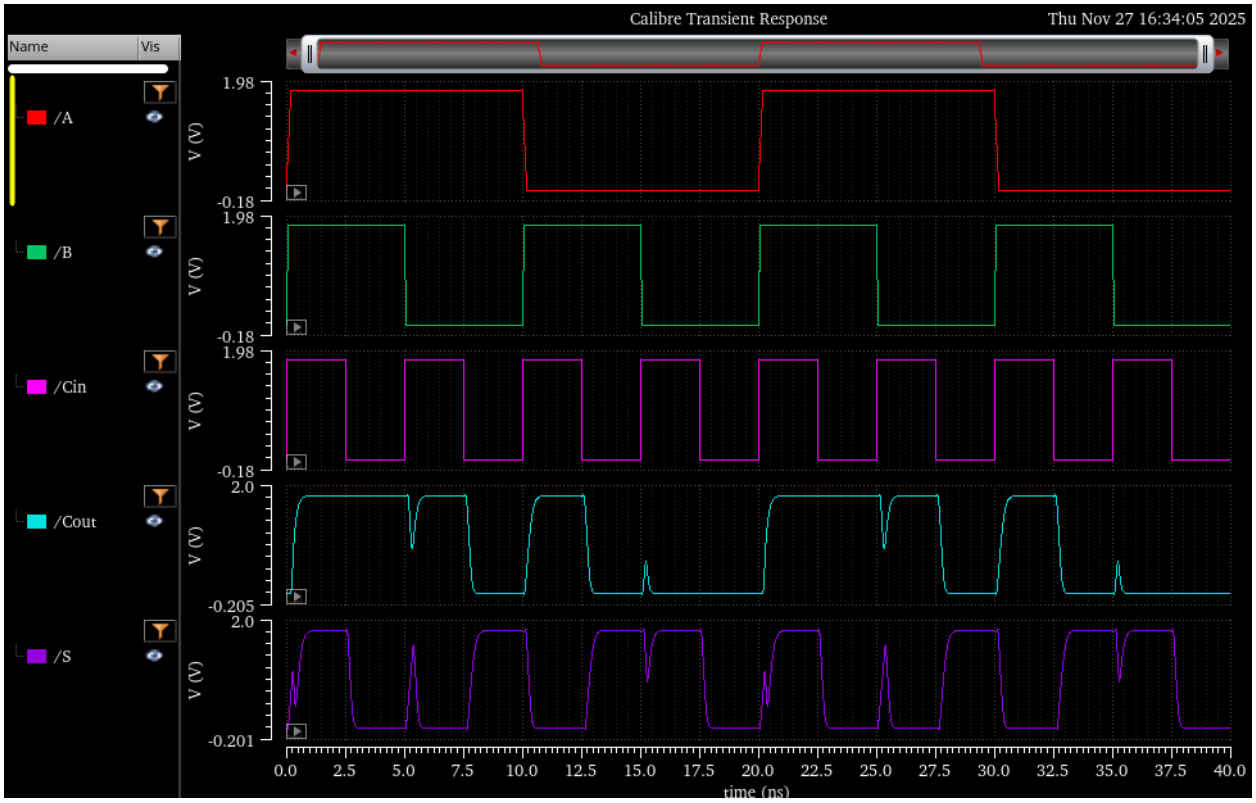


Figure 2.3: Calibre Transient Response

6	Cout Rising Delay	386.768p
7	Cout Falling Delay	363.874p
8	S Falling Delay	385.19p
9	S Rising Delay	458.199p

Figure 2.4: Calibre Rise Fall Delay

Conclusion

In conclusion, the transient response of the spectre matches the transient response of the calibre simulation. The rise and fall delay of both the S and Cout is around the same number with the Calibre simulation being slightly higher by around ~50 pSec. Overall the results were expected with no major deviations.